

# Certus: literature review and roadmap

Based on 19 papers in project knowledge and 3 source repositories, read August 2026

Confidence tags follow the project convention. **[Certain]** means I can point to the specific document text or code in this session. **[Likely]** means published evidence exists but I did not verify it here. **[Guessing]** means plausible from mechanism, unverified.

## 0. Three corrections before anything else

### 0.1 Two files in project knowledge are the wrong papers. **[Certain]**

- [Li\\_et\\_al\\_2025\\_FEABench...](#) is **FEA-Bench** (arXiv 2503.06680, Microsoft Research Asia and Peking University). It is a software engineering benchmark about implementing new features in GitHub repositories from pull requests. It has nothing to do with finite elements. The paper you wanted is **FEABench** by Mudur, Cui, Venugopalan, Raccuglia, Brenner and Norgaard, arXiv 2504.06260, about COMSOL multiphysics. I did read the real FEABench through its source repository, which you uploaded, so the analysis below is not affected.
- [Wiesendanger\\_2024\\_Schwarzschild\\_Solution...](#) is a gauge theory of gravitation paper. It is unrelated to this project. Remove it so it stops consuming project knowledge budget.
- [Guo\\_2026\\_AIempowered\\_Next\\_Generation\\_Computer-Aided\\_Engineering](#) is Jiachen Guo's Northwestern PhD dissertation, not the CMAME journal article. The CMAME article is present separately as [Guo\\_et\\_al\\_2025\\_Large\\_language\\_model-empowered...](#). Both are useful; the labels are just wrong.

### 0.2 My earlier position on the gap needs one more narrowing. **[Certain]**

I previously wrote that "no published system performs runtime detection of a modelling error that runs cleanly and returns plausible numbers." After reading ALL-FEM and Tian and Zhang in full, that statement survives, but it is now much better supported and can be stated more sharply, because two independent groups have now *named the gap themselves in their own words*:

- ALL-FEM (Purdue, CMAME 457:118985, 2026): "a compilable code does not guarantee code correctness. Since the LLM receives no feedback on the numerical approach, it cannot correct formulation or implementation errors that still produce executable code."
- Tian and Zhang (Canterbury, arXiv 2408.13406v2): they name "a verification-validation gap where executable but physically incorrect code passed undetected" as one of three systematic failure modes, and add "No agent combination successfully validated constitutive relations in complex tasks."

This is now the strongest form of the Certus thesis: **the gap is not one I discovered and others missed. It is one that competent groups have identified, published, and then declined to close.** That is a better story than novelty, because it is checkable and because it explains why the gap persists: closing it is hard, and closing it lowers your measured success rate (see 3.4).

**0.3 The most important single finding in this review is not about a competitor.** It is PDE-Agents' own statement of why their measurement works: "The narrow scope is also what makes our central measurement possible. Because the agent fills in parameters for a fixed, independently verified solver instead of emitting solver code, every run produces a configuration whose material properties can be checked directly against ground truth... A system that generates solver code

covers far broader physics but cannot isolate property fidelity the same way, since a failure may lie in the code, in the parameters, or in both." [Certain]

Certus generates decks over free geometry. By their argument, Certus cannot measure itself the way they measure themselves. **This is a direct constraint on Milestone B and it invalidates the obvious benchmark design.** Section 6 gives the way around it.

## 1. What the field actually is, as of August 2026

Nineteen papers, one scoping review, three source repositories. The picture is consistent enough to state in one paragraph.

**Every published agentic FEA or CFD system optimises the same objective: get a simulation to run and finish.** The metrics are executability, schema validity, job lifecycle completion, and percentage of benchmark problems that produce output. Where correctness is measured at all, it is measured **offline, by a human or an LLM judge, against a reference solution that the researchers prepared in advance.** No published system carries a correctness judgement into a run on a problem where no reference exists. [Certain, across all systems reviewed]

The scoping review confirms this is the field's own view of itself. Baker, Rafferty and Price (Queen's University Belfast, *Big Data and Cognitive Computing* 9(12):305, 2025) screened four databases under PRISMA-ScR and analysed 66 studies. Their finding: the most-cited technical challenge is weak spatial and geometric reasoning (16 studies), followed by reliability and factual correctness (12 studies). Their conclusion on evaluation is worth memorising verbatim for the WBSO filing: "Current methods often rely on general NLP metrics or geometric similarity scores that fail to capture the functional and physical validity of a design. Future work should focus on creating comprehensive benchmarks that test for manufacturability, physical plausibility, and adherence to engineering principles." [Certain]

That is a peer-reviewed systematic review of the whole field saying the thing Certus is built on. It is a better citation than any individual competitor paper.

### 1.1 Five clusters

**Cluster A: end-to-end pipeline automation.** FeaGPT (Stuttgart/Exeter, CalculiX), MooseAgent (Nuclear Power Institute of China, MOOSE), Foam-Agent 2.0 (RPI, OpenFOAM), VFEAgent (Peking, Abaqus), Geng et al. (Miami, OpenSeesPy). Goal: natural language to a completed run. Metric: executable success rate. Foam-Agent 2.0 reports 88.2% with Claude 3.5 Sonnet against 55.5% for MetaOpenFOAM, and states plainly that "the success rate is measured by the percentage of cases that ran successfully." MooseAgent reports 93% and its failure analysis is entirely about infinite retry loops. [Certain]

**Cluster B: model specialisation.** ALL-FEM (Purdue). Fine-tunes open-weight models from 3B to 120B on a 1000-entry FEniCS corpus built by a multi-LLM augmentation pipeline from a 503-entry expert seed. Best result 71.79% code-level success on 39 expert-designed benchmarks, beating non-agentic GPT-5 Thinking. This is the most serious engineering effort in the set. [Certain]

**Cluster C: benchmarks.** FEM-Bench (Boston University, arXiv 2512.20732), FEABench (Google DeepMind, COMSOL), ENGDESIGN (UIUC + Amazon AGI, NeurIPS 2025 D&B), ALL-FEM's 39-problem set, Geng's 20 frames, Foam-Agent's 110 tasks. See section 6.

**Cluster D: the vision papers.** Guo, Park, Qian, Hughes and Liu, "Large language model-empowered next-generation computer-aided engineering," CMAME 450:118591 (2026). This is the field-defining paper by the two biggest names in computational mechanics. **It is not about setup verification at all.** It is about using LLMs to automate the derivation and implementation of intrusive model order reduction solvers (TAPS), so that a hard-to-write class of solver becomes accessible. Their hallucination concern is about LLMs deriving wrong weak forms, and their fix is chain-of-thought prompting with curated examples. [Certain]

This matters strategically in two ways. First, Certus is not competing with Hughes and Liu; they are solving the solver-development bottleneck, Certus is solving the setup-correctness bottleneck. Second, and less comfortably: **the most authoritative vision statement of the field does not consider setup verification worth a section.** You should have an answer ready for why they are wrong, or why they are right and it just is not their problem. My read is the latter: they write for researchers who build their own solvers and already know what plane strain means. Certus is for people who do not. [Guessing]

**Cluster E: agentic method discovery.** AgenticSciML (Jiang and Karniadakis, npj Artificial Intelligence 2026) and ATHENA (Toscano, Chen and Karniadakis, Brown). Agents that invent and evolve numerical methods. ATHENA "diagnoses ill-posed formulations" and claims to match or exceed expert-authored reference solutions on canonical benchmarks. Adjacent, not competing: they work on mathematically posed problems, not on a CAD part with an ambiguous boundary condition. Worth watching because Karniadakis's group moves fast and could turn this machinery toward FE setup. [Certain on scope, Guessing on the risk]

## 1.2 One item outside all clusters

Ghorbani et al., "Parameters in play: AlphaZero-inspired AI for autonomous parameter identification in soil constitutive and finite element models," *Computers and Geotechnics* 174:106657 (2024). Deep learning for initial estimates plus Monte Carlo Tree Search for refinement, applied to calibrating advanced soil constitutive models and to inverse FE analyses. No LLM anywhere. [Certain]

Relevance: it is the only paper in the set that automates a *judgement* about material model parameters rather than automating a *workflow*. If Certus ever adds a material-model-validity check that does more than range-testing, this is the prior art to read properly. Not urgent.

## 2. The verification ladder

To compare these systems honestly I needed a scale. Nine levels, ordered by how hard the check is to fool. Levels 0 to 3 are cheap and widely implemented. Levels 4 and up are where Certus lives.

| Level                             | What it checks                                                      | Fails when                                                                   |
|-----------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>L0 Execution</b>               | The solver started and exited without error                         | The job crashes                                                              |
| <b>L1 Schema</b>                  | The input file parses, required fields present, types correct       | Missing keyword, bad type                                                    |
| <b>L2 API existence</b>           | Every function or keyword used actually exists in the target tool   | Hallucinated API call                                                        |
| <b>L3 Parameter range</b>         | Values are positive, finite, inside a plausible band                | Negative Poisson's ratio, $k$ less than 0                                    |
| <b>L4 Input fidelity</b>          | The model reflects what the user actually specified                 | BC applied to the wrong face; dimension dropped                              |
| <b>L5 Discretization adequacy</b> | The element formulation and mesh can represent this response        | Shear locking, volumetric locking, hourglassing, unresolved wall             |
| <b>L6 Idealization validity</b>   | The modelling abstraction matches the physics being asked for       | Plane strain used for plane stress; elastic model past yield; overconstraint |
| <b>L7 Post-solve closure</b>      | The result is consistent with the assumptions that produced it      | Stress exceeds yield in a linear-elastic run and nobody notices              |
| <b>L8 Refusal</b>                 | The system declines when no valid configuration exists on its stack | Answers anyway                                                               |

Two properties matter for the pitch. **L0 to L3 catch errors that announce themselves.** L4 to L7 catch errors that run cleanly, converge, and return plausible numbers. **L8 is not a check at all; it is a policy**, and it is the one every published system explicitly designs against.

## 2.1 Every system against the ladder

| System                | L0        | L1  | L2  | L3  | L4           | L5                               | L6                         | L7                | L8          |
|-----------------------|-----------|-----|-----|-----|--------------|----------------------------------|----------------------------|-------------------|-------------|
| FeaGPT                | yes       | yes | —   | yes | no           | partial (Jacobian, aspect ratio) | no                         | no                | no          |
| MooseAgent            | yes       | yes | yes | no  | no           | no                               | no                         | no                | no          |
| Foam-Agent 2.0        | yes       | yes | yes | no  | no           | no                               | no                         | no                | no          |
| VFEAgent              | yes       | yes | yes | yes | yes          | no                               | no                         | no                | <b>anti</b> |
| Geng et al.           | yes       | yes | yes | no  | partial      | no                               | no                         | no                | no          |
| ALL-FEM               | yes       | yes | yes | no  | offline only | no                               | offline only               | no                | no          |
| PDE-Agents            | yes       | yes | yes | yes | no           | partial (fixed thresholds)       | no                         | partial (Tscore)  | <b>anti</b> |
| Tian and Zhang        | yes       | yes | —   | no  | no           | no                               | <b>measured and failed</b> | no                | no          |
| FEABench              | yes       | yes | yes | no  | no           | no                               | no                         | target value only | no          |
| FEM-Bench             | yes       | yes | —   | —   | —            | —                                | code-level only            | —                 | —           |
| <b>Certus (today)</b> | <b>no</b> | yes | —   | —   | yes          | designed, one rule defective     | partial                    | designed          | designed    |

"anti" means the system contains a deliberate mechanism to prevent refusal. VFEAgent has a deterministic fallback whose stated purpose is to guarantee an answer is always produced. PDE-Agents moved its pre-run checks off the mandatory path because enforcing them cost success rate.

The **no** in Certus's L0 column is the whole problem. Every column to the right of it is unverifiable until it is fixed.

## 2.2 What the checks actually are, in code

I read the three repositories rather than trusting the papers. [Certain, from source]

**FeaGPT.** `feagpt/geometry/validators.py` implements a three-layer validator described as "syntax, physics, and manufacturability." Layer 2, the "physics" layer, checks that dimension fields are numeric, positive, above 0.1 mm and below 10000 mm. Layer 1 checks the geometry type against a hard-coded whitelist of eight strings: cantilever-beam, naca\_wing, plate\_with\_hole, bracket, I-beam, tube and two aliases. `feagpt/meshing/quality.py` computes aspect ratio and a Jacobian-based quality metric with a fixed threshold of 0.3 and max aspect ratio 10.

**The finding that matters most is in `feagpt/meshing/mesher.py`.** The function `_create_physical_groups` assigns boundary conditions to faces **by enumeration index**:

```

for i, bc in enumerate(bcs):
    loc = bc.get("location", f"bc_{i}").replace(" ", "_")
    if i < len(surfaces):
        gmsh.model.addPhysicalGroup(2, [surfaces[i][1]], name=loc)
for i, load in enumerate(loads):
    idx = len(bcs) + i
    if idx < len(surfaces):
        gmsh.model.addPhysicalGroup(2, [surfaces[idx][1]], name=loc)

```

The semantic location string, which the paper describes as intelligently inferred ("left edge", "hole boundary"), is used **only as a label**. The face is chosen by whatever order Gmsh happened to enumerate surfaces in. Separately, `feagpt/simulation/simulator.py` writes `*BOUNDARY` referencing a node set called `NFIX` and `*CLOAD` referencing `NLOAD`, and neither set is defined anywhere in the repository. **[Certain]**

**Caveat, and it is important.** This repository's LICENSE reads "Copyright (c) 2026 naividh," which does not match the paper's authors (Qi, Xu, Chu). I cannot confirm this is the authors' own code. What I can say is that the leading public artefact called FeaGPT selects boundary faces by index order. If it is their code, the paper's central claim of "proper boundary condition inference" is not supported by the implementation. If it is not their code, someone has published a plausible-looking reimplementaion that silently attaches loads to arbitrary faces, which is itself a perfect illustration of the failure mode Certus targets.

**MooseAgent.** `src/mooseagent/utlis.py` contains exactly one deterministic check, `check_app`. It extracts every `type = <Name>` from the input card and reports any name absent from the documentation JSON. That is L2 and nothing else. The rest of the loop is LLM error-log interpretation and retry, with a human-intervention trigger after three repeats of the same error signature. **[Certain]**

**FEABench.** `common/eval/api_score.py` defines `physics_code_metrics`. The metrics named "interface\_realism," "interface\_code\_recall," "feature\_code\_recall" and "correct\_dimension\_features" are **string and API-call recall against ground-truth COMSOL Java code**. `evaluate_bench.py` scores on Executability, ExportedValue and TargetAccuracy, where TargetAccuracy is relative error against a known target value taken from a COMSOL tutorial. The "physics metrics" contain no physics. They measure whether the model called the same COMSOL API functions the tutorial called. And the whole harness requires a COMSOL licence through an MPh client. **[Certain]**

## 3. Six findings that change how Certus should be built

### 3.1 The one directly comparable experiment says LLM verification does not work

Tian and Zhang ran 1,120 controlled trials, seven role configurations, four linear-elastic FEA tasks, 12-turn limit. Findings: **[Certain]**

- The Rebuttal agent, explicitly prompted to be adversarial, agreed with the Critic 85 to 92% of the time, including when the Critic was wrong. In Task 1, 39 of 43 Rebuttal interventions

(90.7%) were affirmations. Of 52 Critic interventions in failed cases, 16 (30.8%) contained explicitly incorrect diagnoses, and the Rebuttal agent affirmed them anyway. Fisher's exact test across configurations  $p = 1.000$ , so this is not a configuration artefact.

- Adding a redundant reviewer **reduced** success.
- Only the three-agent Coder-Executor-Critic configuration produced physically and visually correct solutions.
- Their third named failure mode is the verification-validation gap: executable but physically incorrect code passing undetected.

**Consequence for Certus:** the architectural rule already in the project instructions, that physics decisions are deterministic and LLM calls are bounded to three narrow edges, is now backed by 1,120 trials from an independent group. Do not weaken it. When a funded competitor ships "an AI reviewer agent," this is the paper that says why it will not work.

### 3.2 The most rigorous group in the field made their physics checks optional, and said why

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PDE-Agents (MatPro, Institute for Energy Technology, Kjeller). Their `rules.py` fires nine pre-run checks. Their KG On mode makes those checks mandatory before every run: 47 of 50 tasks succeed (94%). Their KG Smart mode makes them lazy, used only on the failure-recovery path: 50 of 50 (100%), with higher output quality too (physics score 0.933 vs 0.853, MPF 0.926 vs 0.796).

[Certain]

Their diagnosis, verbatim: "The `check_config_warnings` tool occasionally returns pre-emptive warnings (e.g. 'CFL criterion may be violated') that the model interprets as prohibitive, leading it to report a warning instead of proceeding. **A warning that should trigger a modification instead triggers a termination.**" [Certain]

**This is the single best argument Certus has, and it is not the argument I would have chosen.** The naive pitch is "nobody checks physics." The true and much stronger pitch is: *the one group that built real deterministic physics checks measured what enforcing them costs, found it costs success rate, and responded by taking them off the critical path.* That is a structural incentive problem in the whole field, not an oversight by one team, and it is exactly the kind of problem a company can be built on.

**Consequence for Certus:** the product's hardest engineering problem is not detecting the error. It is making the finding **actionable rather than terminal**. A Certus finding must carry a fix owner and a proposed lever, so the pipeline modifies instead of stopping. The `cure_availability()` function in `solvers.py` is already the right shape for this. It should be treated as a headline feature, not plumbing.

### 3.3 The unobservable-error trap, and the second trap underneath it

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PDE-Agents' Table 6 shows steady-state Dirichlet cases where 86%, 64% and 100% material-property errors produce **zero** output error, because those solutions are analytically independent of  $k$ ,  $\rho$  and  $c_p$ . [Certain]

First consequence, already recorded: seed only errors that provably propagate to the quantity of interest, and report the magnitude, or the false-negative rate is meaningless.

Second consequence, new and worse, from their limitations section quoted in 0.3: their measurement works *because* the solver is fixed and independently verified, so a failure can only be

in the parameters. Certus generates decks over free geometry, so a bad result could come from the geometry, the mesh, the deck, or the check. **Certus cannot copy their benchmark design.**

The way out is to fix everything except the one decision under test. See 6.3.

### 3.4 The strongest concrete example of a silent error in the literature is ALL-FEM's, not mine

ALL-FEM's correctness evaluation is a three-step offline process: discard non-executable code; plot the candidate and the reference together and call them equal if indistinguishable at plot scale; then run an **LLM judge** over the code diff and manually verify what it flags. Their worked example: Llama 3.3 70B in the two-agent framework produced "a solution plot indistinguishable from the reference," and the code-level analysis revealed it had used a **plane strain formulation where the reference required plane stress**. Marked incorrect. [Certain]

Three things follow.

1. This is a textbook L6 idealization error, caught only because a human had a reference solution and an LLM diffed the code against it. On a real user problem there is no reference to diff against.
2. Their stated reason for not using relative L2 error is that different frameworks produce solutions on different meshes, and projecting between them is hard to automate robustly. This is the same reproducibility problem already recorded in the Certus engineering principles (84,584 elements on Linux vs 84,600 on Windows). Independent confirmation that **benchmarks must assert on the finding, never on a raw number**.
3. Plane stress versus plane strain should be a named Certus check. It is the cleanest possible demo case: two-line change, indistinguishable plot, wrong answer, and a senior engineer catches it in five seconds. Put it in the negative benchmark.

### 3.5 LLMs are very bad at mechanical design when the grader is a simulator

ENGDESIGN (NeurIPS 2025 Datasets and Benchmarks) replaces answer-checking with simulation-driven functional verification across nine domains. Average pass rate, Mechanical Systems column, three trials per task: [Certain]

| Model             | Mech         | Struct       | Overall |
|-------------------|--------------|--------------|---------|
| GPT-4o            | 0.00         | 25.64        | 15.68   |
| DeepSeek-v3       | 0.00         | 38.89        | 17.92   |
| Gemini-2.0-Flash  | 4.76         | 36.11        | 14.16   |
| Claude-3.7-Sonnet | 14.29        | 30.77        | 22.61   |
| o1                | 14.29        | 23.08        | 29.17   |
| o3                | 23.81        | 30.77        | 34.38   |
| o3-high           | <b>28.57</b> | 41.03        | 33.57   |
| Gemini-2.5-Pro    | 0.00         | <b>50.00</b> | 29.54   |
| DeepSeek-R1       | 4.76         | 41.67        | 25.53   |

The best score any model achieves on mechanical systems is 28.57%. Gemini-2.5-Pro scores zero.

Use this carefully. It measures *design* tasks, not FE setup, and the models are a generation old. It does not prove that a 2026 frontier model cannot set up a bracket analysis. What it does support, defensibly, is the narrower claim that **when the grader is a simulator rather than a text comparison, mechanical-domain performance collapses**, and that this collapse is domain-specific rather than general (the same models score 40 to 69% on robotics and computer architecture). That is enough for a slide. Do not stretch it further.

### 3.6 The benchmark most likely to eat Milestone B has now stated its roadmap

FEM-Bench (Boston University, Lejeune group, arXiv 2512.20732v2, updated 29 May 2026). 33 tasks from a first graduate course in computational mechanics. Best function-writing model Gemini 3 Pro completes 30/33 at least once in five attempts, 26/33 five out of five. Best test-writing model GPT-5 scores 73.8% Average Joint Success Rate. [Certain]

Two things make it directly relevant.

**It already contains a negative benchmark, at a different layer than Certus's.** Each task ships "expected failure implementations": manually written known-incorrect functions that a good unit test must detect. Their categories are formula errors (swapping  $I_y$  and  $I_z$ ), sign or ordering errors, missing physics (dropped geometric stiffness terms), missing numerical checks (solving without ill-conditioning detection), and trivially wrong outputs. They state the reason plainly: ensuring failure on known-bad implementations makes joint test success "a meaningful check, rather than an incidental property of the output." [Certain]

**That is the FEM-Bench team independently arriving at the Certus engineering principle that a check which cannot fail is worse than no check.** Cite it.

**And their stated future work is on a collision course:** "nonlinear material behavior, dynamic analysis, incompressible elasticity and associated numerical challenges, and multiphysics coupling." Incompressible elasticity is volumetric locking territory. [Certain]

But the layers are different, and this is the distinction that protects Certus. **FEM-Bench asks whether the LLM can correctly implement an element routine. Certus asks whether a correctly implemented solver was correctly applied to a real part.** FEM-Bench's expected failures are bugs in the code. Certus's are bugs in the modelling decision. They are complements, not competitors, and the right relationship is to build on their harness design rather than duplicate it.

## 4. Honest inventory: what they have that Certus does not

I ranked these by how much they should worry you.

**1. A working solve. All of them.** Certus has never executed a solver. This is not a feature gap, it is a credibility gap, and every claim in sections 2 and 3 above is unverifiable from the Certus side until it closes.

**2. Public benchmarks with released data.** ALL-FEM's 39 expert-designed problems with reference solutions, and PDE-Agents explicitly names it as their next test set. FEM-Bench's 33 tasks with a fully automated re-runnable pipeline. FEABench Gold and FEABench Large (200 problems). ENGDESIGN on HuggingFace. Geng's 20 frames. Foam-Agent's 110 tasks. Certus has zero published test cases.

**3. Fine-tuned domain models.** ALL-FEM built a 503-entry expert seed, expanded it to 1005 entries through a multi-model augmentation pipeline with domain-expert validation at every step, and LoRA/QLoRA fine-tuned models from 3B to 120B. That is roughly a person-year of graduate labour that Certus does not have and should not try to replicate.

**4. Knowledge bases.** MooseAgent ships a 230 MB vector database of annotated MOOSE input cards and function documentation. PDE-Agents ships a Neo4j GraphRAG with material nodes, failure-pattern nodes and run lineage. Certus has none, and the honest question is whether it needs one.

**5. Multimodal input.** VFEAgent parses engineering drawings to Abaqus with 90.0% schema validity against 41.7% for the best VLM baseline. Certus's image-to-CAD path exists but is untested at that level.

**6. Downstream product surface.** FeaGPT ships fatigue analysis, Pareto optimisation, sensitivity analysis, surrogate models, batch parameter sweeps (432 NACA configurations), a REST API and a Cloud Run deployment script. Foam-Agent 2.0 exposes itself as MCP tools so Claude Code can drive it, generates HPC submission scripts, and does ParaView post-processing. Certus writes a deck.

**7. Published formal V&V.** PDE-Agents demonstrates  $O(h^2)$  convergence, rates 2.04 and 2.00 against closed-form solutions, anchored to ASME V&V 10-2006.

**8. Teams and institutions.** Purdue (Gomez, Vlachos), Brown (Karniadakis), Peking, Northwestern (Liu, with Hughes at UT Austin), Google DeepMind, IFE Norway, Nuclear Power Institute of China. Certus is one person with one to two days a week.

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## 5. What Certus has that they do not

Each of these is verified against source code or paper text, not assumed. [Certain unless marked]

**1. Deterministic face selection on arbitrary CAD, with a confirmation step.** This is the biggest one and I had underrated it. Look at how each system gets a boundary condition onto a surface:

- FeaGPT: whitelist of 8 parametric geometry types, faces chosen by enumeration index.
- VFEAgent: 2D orthographic drawings, extruded profiles. Their own limitations section says single-view 3D reconstruction is ill-posed and complex non-manifold geometry needs multi-view reasoning they do not have.
- ALL-FEM and PDE-Agents: canonical domains defined analytically in code. Unit squares, rectangles, a Turek-Hron beam.
- MooseAgent: the user supplies a mesh file *and tells the agent the names of the boundaries in it*.
- Geng et al.: 2D frames, node coordinates derived by expert rules from bay and storey counts.

**Not one of them takes an arbitrary STEP file and works out which face is the pin hole.** Certus's `geometry_features.py` face catalogue plus `geom_session.py` single-session tag namespace does exactly that, verified on a real bracket producing 27 feature groups, with `PIN_HOLE` confirmed cylindrical about X at diameter 8.0000 with 0.000% radius spread and `MOUNT_FACE` confirmed planar normal to Z. This is a genuine technical asset and it has been sitting in the "plumbing" mental category when it should be in the "differentiator" category.

**2. Locking rules tied to load regime.** No system in the set has any L5 check beyond mesh quality metrics with fixed thresholds. PDE-Agents' mesh rules are `nx < 10` in 2D and `< 8` per direction in 3D, geometry-blind and solution-blind. FeaGPT's are aspect ratio 10 and Jacobian quality 0.3. `locking_check.py`'s eight rules with a severity ladder and owner routing have no counterpart.

**3. Solver capability routing and refusal.** `cure_availability()` turning a finding into a stack decision has no counterpart. [Likely, since I checked the reviewed set but cannot rule out an unreviewed system]. Both systems that address refusal at all address it in the opposite direction: VFEAgent's deterministic fallback exists to guarantee an answer; PDE-Agents treats a pre-run warning that stops a run as a failure mode to be engineered away.

**4. Independent read-back verification.** `verify_sets.py` reads the written deck and answers a geometric question about the actual node coordinates, independent of Gmsh and of the solver. Every other system verifies through the tool that will consume the artefact, or against a reference the researchers wrote. ALL-FEM's LLM judge reads the code, not the model.

**5. Check-provenance guarding.** Nobody discusses it. Certus has the concept and currently violates it (R7). That is embarrassing internally and valuable externally: it is a named failure class with a worked example, which is exactly what a WBSO technical-risk narrative needs.

**6. Cross-platform verdict stability as a stated requirement.** ALL-FEM ran into the underlying problem and worked around it by not using L2 error. Certus states it as a design constraint on the benchmark.

## 6. Answers to your specific questions

### 6.1 Do these tools avoid commercial solvers?

Mostly yes, and this removes "we are the open-source one" as a differentiator. [Certain]

| System                | Solver                     | Commercial licence needed |
|-----------------------|----------------------------|---------------------------|
| ALL-FEM               | FEniCS/DOLFINx             | no                        |
| PDE-Agents            | FEniCSx/DOLFINx            | no                        |
| FeaGPT                | CalculiX                   | no                        |
| MooseAgent            | MOOSE                      | no                        |
| Foam-Agent 2.0        | OpenFOAM                   | no                        |
| Geng et al.           | OpenSeesPy                 | no                        |
| MechAgents            | FEniCS                     | no                        |
| AutoFEA               | CalculiX                   | no                        |
| <b>VFEAgent</b>       | <b>Abaqus</b>              | <b>yes</b>                |
| <b>FEABench</b>       | <b>COMSOL</b> (MPh client) | <b>yes</b>                |
| SimPilot (commercial) | OpenFOAM, SU2, CalculiX    | no                        |

Open-source solvers are the field default. **Do not put "no commercial licence required" on a slide as a differentiator.** It is table stakes. It remains useful as a *cost* argument to an SME that currently pays for seats, but that is a sales talking point, not positioning, and it is adjacent to the rejected "cheaper simulation" line the project instructions already forbid.

One genuinely useful consequence: **CalculiX is the established open-source structural choice** (FeaGPT, AutoFEA, SimPilot). The stack decision in open question 3 is therefore consistent with the field, and its known limitation, no hybrid elements at all, is a real and citable reason for the capability-routing feature rather than an embarrassment.

## 6.2 Do they run on local LLMs?

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Several do, and the trend is clearly toward open-weight models. [Certain]

- **ALL-FEM:** open-weight only, fine-tuned. Llama 3.2 3B, Llama 3.3 70B, Qwen 3 32B, GPT-OSS 120B. Best result 71.79% from a fine-tuned GPT-OSS 120B, beating non-agentic GPT-5 Thinking. This is the strongest evidence in the set that a self-hosted model plus a good agentic framework beats a frontier API model used naively.
- **PDE-Agents:** locally deployed open-source LLMs throughout, with cross-model validation across two generations.
- **Geng et al.:** Llama-3.3 70B Instruct, chosen explicitly "for open-source accessibility and deployment scalability," and it beats Gemini-2.5-Pro and GPT-4o on their benchmark.
- **FEM-Bench:** evaluates open-weight models alongside frontier ones and names fine-tuning open-weight models on its tasks as future work.
- **VFEAgent, FeaGPT, Foam-Agent 2.0, MooseAgent:** frontier APIs (GPT-4 class, Claude 3.5 Sonnet, DeepSeek-R1).

**Consequence for Certus, and it is a real one.** The EU data sovereignty angle in the project instructions is currently listed as unverified but plausible. It is now also **technically achievable at low cost**, because three independent groups have shown a 70B-class open-weight model is sufficient for the LLM roles in this pipeline. And Certus needs *less* from its LLM than any of them, because the deterministic/LLM split confines the model to three narrow edges. If ALL-FEM needs a fine-tuned 120B to write FEniCS code and Certus only needs a model to turn "fix the mounting face" into a catalogue group id, then **Certus can credibly promise a fully on-premise, no-data-leaves-the- building deployment that none of the API-based systems can match.** That converts a soft positioning claim into a hard technical one. It is worth one slide and one discovery-call question.

## 6.3 What must Milestone B actually be

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Constraints, all now evidence-based:

- Cannot assert on raw numbers (ALL-FEM's mesh-projection problem, Certus's own platform variance).
- Cannot seed errors that do not propagate (PDE-Agents Table 6).
- Cannot use a design where failure could lie in geometry, mesh, deck, or check (PDE-Agents' measurability argument).
- Must fail on known-bad inputs by construction (FEM-Bench's expected-failure design, and Certus's own R7 as the counterexample).

The design that satisfies all four: **one fixed part, one fixed load case, one fixed solver, one fixed reference solution, and exactly one modelling decision varied per case.**

Structure per case: 1. A known-good baseline deck that runs and matches an analytical or converged reference. 2. One deliberate modelling change. Element formulation, idealization, constraint set, material regime. 3. A **measured** propagation magnitude: run both, record the error in the quantity of interest. If it is below noise, the case is discarded as unobservable and reported as such. 4. The expected Certus finding: rule id, severity, owner, cure. The assertion is on this, never on a stress value. 5. A matched control where the same rule must **not** fire, so the check cannot pass by always firing.

Twelve cases is enough for a first version. Candidates, ordered by how cleanly they demonstrate the thesis:

1. Plane stress used where plane strain is required (ALL-FEM's own worked failure)
2. Linear fully-integrated hex in bending: shear locking
3. Linear tet in near-incompressible response: volumetric locking, and CalculiX has no cure, so this is also the refusal demo
4. Reduced integration without hourglass control
5. Fully clamped face where a pinned or partial constraint is physical: overconstraint
6. Linear elastic material used past yield
7. Wall thickness resolved by one element through-thickness
8. Load applied to the wrong face, geometrically adjacent and plausible (the FeaGPT failure mode)
9. Symmetry boundary condition on a non-symmetric load
10. Mesh too coarse at a stress concentration, with a convergence gate
11. Rigid-body mode not eliminated, solver returns something anyway
12. Units inconsistency that survives dimensional plausibility checks

Build it on ALL-FEM's public 39-problem set where the physics overlaps, so the baseline is externally validated rather than self-declared, and anchor the language to ASME V&V 10-2006 as PDE-Agents does.

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## 7. Revised gap list

Updated from the nine-row version, with evidence.

| #  | Gap                                                                                                                    | Status after this review                                                                                                                                                                                                                       |
|----|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Discretization-error adjudication: element formulation against load regime; shear and volumetric locking; hourglassing | <b>Unoccupied.</b> Best existing L5 is a fixed-threshold mesh quality metric. FEM-Bench lists incompressible elasticity as future work. Structurally impossible for PDE-Agents (scalar field).                                                 |
| 2  | Solver-capability routing; refusal when no cure exists on the stack                                                    | <b>Unoccupied.</b> Every reviewed system is single-solver.                                                                                                                                                                                     |
| 3  | Overconstraint auditing                                                                                                | <b>Unoccupied.</b> VFEAgent's L2 checks static determinacy for under-constraint only.                                                                                                                                                          |
| 4  | Idealization validity (plane stress vs plane strain, symmetry, dimensional reduction)                                  | <b>Unoccupied at runtime.</b> ALL-FEM catches it offline with an LLM judge plus a human plus a reference solution.                                                                                                                             |
| 5  | Input-fidelity: the BC is on the face the user meant                                                                   | <b>Weakly occupied.</b> VFEAgent L4 reverse-checks drawing dimensions against extracted topology. Nobody does it on arbitrary 3D CAD. FeaGPT's public code assigns by index.                                                                   |
| 6  | Post-solve assumption closure                                                                                          | <b>Weakly occupied.</b> PDE-Agents' Tscore checks Tmax/Tmin against expected ranges.                                                                                                                                                           |
| 7  | Mesh convergence as a per-run gate                                                                                     | <b>Partly occupied.</b> PDE-Agents does h-refinement offline once, for solver verification. Nobody does it per-run per-part.                                                                                                                   |
| 8  | Refusal as a valid output                                                                                              | <b>Unoccupied, and now strongly evidenced as actively designed against.</b>                                                                                                                                                                    |
| 9  | Check-provenance guarding                                                                                              | <b>Unoccupied and undiscussed anywhere.</b>                                                                                                                                                                                                    |
| 10 | Verdict reproducibility across platforms                                                                               | <b>Unoccupied.</b> ALL-FEM hit the underlying problem and routed around it.                                                                                                                                                                    |
| 11 | Negative benchmark for discretization and idealization errors in structural mechanics                                  | <b>Unoccupied.</b> FEM-Bench has expected-failure implementations at the <i>code</i> layer. PDE-Agents has a propagation-aware methodology for material-property errors on a scalar solver. Neither covers modelling decisions on a real part. |

Rows 1, 2, 3, 4, 8, 9, 10 and 11 are unoccupied. That is a wider opening than the previous assessment, mostly because reading the code showed that rows 4 and 5 are weaker in practice than the papers claim.

## 8. The strategic fork this review opens

I did not expect to find this, and it is the item I would most want you to push back on.

Certus is roughly two years behind on the generation layer and cannot catch up solo. ALL-FEM has a fine-tuned model stack. Foam-Agent has MCP tooling and HPC integration. FeaGPT has optimisation loops and a deployed API. Certus has a face catalogue and no solver run.

But **the verification layer does not need to own the pipeline.** `verify_sets.py` already demonstrates the shape: it reads a written deck and answers a geometric question about it,

independent of the tool that produced it. A checker that ingests a deck plus a stated intent and returns findings with severity, owner and cure is:

- Buildable by one person, because it does not need CAD generation, meshing UX, or a solver runner
- Testable against artefacts produced by other people's systems, including ALL-FEM's public benchmark
- Complementary rather than competitive, which makes Milestone C conversations easier
- Attachable to SimPilot, SimScale, or an internal Ansys workflow rather than displacing them
- Aligned with the one thing Certus has that nobody else does

The counter-argument is real. A checker is a feature, and features get bought or copied, whereas an end-to-end product is a company. It also has no natural way to charge, and it puts Certus downstream of somebody else's roadmap.

**I am not recommending the fork. I am recommending you decide it deliberately before Milestone D rather than by drift.** The one thing that resolves it is Milestone C: ask ten people whether they would pay for a checker that audits the FE model they already have. If the answer is yes, the solo-founder execution risk in the project instructions drops sharply.

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## 9. Roadmap

Milestone-gated. Nothing here is date-gated. The ordering is the point.

### Phase 0 — Close Milestone A. Nothing else first.

Nothing in this review changes the ranking in section 9 of the project instructions, and everything in it makes the ranking more urgent, because every differentiator claimed above is unverifiable without a solve.

1. Load and constraint writer. `*MATERIAL` , `*SOLID SECTION` , `*STEP` , `*BOUNDARY` , `*CLOAD` .
2. CalculiX run on the bracket.
3. Comparison to a hand calculation with a stated tolerance band.

**Explicit warning per section 12 of the project instructions.** Everything in sections 4, 5 and 8 above will tempt you toward another upstream layer: a knowledge base like MooseAgent's, drawing parsing like VFEAgent's, fine-tuning like ALL-FEM's. Every one of those is the failure mode this project has already demonstrated once. **Build none of them until a solver has run.**

### Phase 1 — Make the checks trustworthy

1. **Fix R7.** Tag `char_size` with provenance and make the rule return `NOT_EVALUATED` when its input derives from its own output. This is now more than a bug fix: check-provenance guarding is gap row 9, unoccupied and undiscussed in the entire literature, and R7 is the worked example that makes it credible.
2. **Compute `dominant_mode`** from load resultant against constraint centroid. Every locking rule depends on it and it is currently asserted from the command line.

3. **Overconstraint rule.** `MOUNT_FACE` is 12.6% of nodes on the real bracket, fully clamped. This will corrupt the Milestone A analytical comparison, so it is not optional. Gap row 3.
4. **Provenance tagging as a general mechanism**, not a special case for R7. Every rule input carries where it came from; any rule whose input traces back to its own output abstains.

## Phase 2 — Milestone B, designed to the constraints in 6.3

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1. Freeze the finding schema: rule id, severity, owner, cure, propagation magnitude, provenance.
2. Build 12 negative cases per the structure in 6.3, each with a measured propagation magnitude and a matched control.
3. Assert on findings only, never on stress values.
4. Where physics overlaps, build on ALL-FEM's public 39-problem set. Anchor language to ASME V&V 10-2006.
5. Add the actionability layer that PDE-Agents' result demands: every finding proposes a modification, and refusal fires only when `cure_availability()` returns nothing on the stack.

## Phase 3 — Milestone C, starting now, in parallel

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Not gated on A or B. Zero customer conversations remains the largest strategic risk in the project.

1. Write the one-page problem statement. It now has four external citations that make the problem real without a demo: Tian and Zhang's verification-validation gap, ALL-FEM's plane stress example, PDE-Agents' warning-induced conservatism, Baker's review conclusion on benchmarks.
2. Pick one segment. Open question 1 is still open and still blocking.
3. Ten conversations, written notes after each, per the project instructions' question list, plus two new ones: (a) would you pay for a checker that audits a model you already built, versus a tool that builds it for you; (b) does on-premise, no-data-leaves-the-building matter enough to change your vendor choice.
4. **Contact the MatPro group at IFE Kjeller.** Government institute, open source, EU, and they name vector mechanics as the extension they are not taking. No working solve required for that conversation.

## Phase 4 — Only after A, B and C

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1. WBSO filing. The technical-risk narrative now writes itself and is externally corroborated: reliable automated detection of finite element modelling errors that run cleanly and return plausible results, validated against a negative benchmark of known-good and known-bad models, with cited evidence that existing systems measure only executability and that the one group that enforced physics checks measured a reliability penalty and removed them.
  2. Lawyer check on Article 11 before incorporation. Unchanged, still required.
  3. KVK, EUIPO trademark, domain. Unchanged.
  4. University incubator as participant, not spin-out. Unchanged.
-

## 10. Citations worth having ready

For the pitch, WBSO and any grant text. All verified against the documents this session. [Certain]

| Claim                                                                                                                        | Source                                                                    |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Executable but physically incorrect code passes undetected; no agent combination validated constitutive relations            | Tian and Zhang, arXiv 2408.13406v2                                        |
| LLM reviewers affirm rather than challenge, 85 to 92%, including errors; redundant reviewers reduce success                  | same                                                                      |
| A compilable code does not guarantee correctness; the LLM gets no feedback on the numerical approach                         | Deotale et al., ALL-FEM, CMAME 457:118985 (2026)                          |
| A plane-strain-for-plane-stress substitution produced a plot indistinguishable from the reference                            | same                                                                      |
| Enforcing pre-run physics checks lowered success from 100% to 94%; the team responded by making them optional                | Adhikari, Noorsumar and Jensen, PDE-Agents, arXiv 2606.07850v2            |
| A warning that should trigger a modification instead triggers a termination                                                  | same                                                                      |
| Property errors of 64 to 100% can produce zero output error where the solution is independent of them                        | same, Table 6                                                             |
| Narrow scope is what makes the measurement possible; a deck-generating system cannot isolate error the same way              | same, limitations                                                         |
| Current evaluation methods fail to capture functional and physical validity; benchmarks are needed for physical plausibility | Baker, Rafferty and Price, <i>Big Data Cogn. Comput.</i> 9(12):305 (2025) |
| Weak spatial and geometric reasoning is the most-cited challenge (16 of 66 studies); reliability second (12)                 | same                                                                      |
| A check that cannot fail is an incidental property, not a meaningful check                                                   | Mohammadzadeh et al., FEM-Bench, arXiv 2512.20732v2                       |
| Best mechanical-systems pass rate across 12 models under simulation-based grading is 28.57%                                  | Guo et al., ENGDESIGN, NeurIPS 2025 D&B                                   |
| Success rate is measured as the percentage of cases that ran successfully                                                    | Yue et al., Foam-Agent 2.0, arXiv 2509.18178v2                            |

Two things **not** to claim, because I checked and they do not hold:

- Do not claim nobody checks anything before running. VFEAgent has four pre-solve validation levels; PDE-Agents has nine deterministic rules. Claim instead that the checks operate at L1 to L3 and that no published system checks discretization adequacy or idealization validity at runtime.
- Do not claim an open-source stack as a differentiator. It is the field default.

## 11. What would falsify the thesis

Stated so it stays checkable rather than becoming an article of faith.

1. **Ten discovery conversations where nobody reports a wrong-but-plausible simulation costing them anything.** The failure mode is real; that it is *painful enough to pay for* is unverified.
2. **A frontier model in 2026 or 2027 reliably catching locking and idealization errors from a deck.** Testable in an afternoon once Milestone A is done: give the bracket deck with a known seeded error to the best available model and see if it catches it. Do this test. If the answer is yes, the moat is the benchmark corpus, not the checks.
3. **SimPilot or SimScale shipping structural setup-correctness checks.** SimPilot already has a V&V stage; its published example is CFD-specific and reference-database-dependent. Re-check quarterly.
4. **Warning-induced conservatism reproducing inside Certus.** If Certus's own checks reduce the rate at which useful answers are produced, PDE-Agents' finding applies to Certus too and the actionability layer is the product, not the checks.
5. **The benchmark not distinguishing Certus from a well-prompted general model.** If a generic agent scores similarly on the 12 negative cases, the differentiator is not where I think it is.

Item 2 is cheap and should be run immediately after Milestone A closes. It is the single highest-value experiment in this document.